Continuous Random Variables and the Normal Distribution

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LESE152

Continuous Random Variables: Several Concepts

- Continuous Random Variables
- The Uniform Distribution
- The Normal Distribution
- The Exponential Distribution
- Multiple Random Variables
- Practice Questions

Continuous Random Variables

- Continuous random variable: A random variable whose possible values fill an interval, so they cannot be listed.
- Examples: a price, a wage, the amount of rainfall, the time it takes to drive to school, the labor force participation rate (any number between 0% and 100%).
- Each single value has probability zero, so the math uses calculus instead of sums.

Cumulative Distribution Function (CDF)

- The CDF has the same definition as in Lecture 5:

$$F(x) = P(X \leq x).$$

- A discrete CDF is a step function. A continuous CDF increases smoothly: it can have flat parts, but it never jumps.
- Standard uniform: X can take any value in $[0, 1]$, with

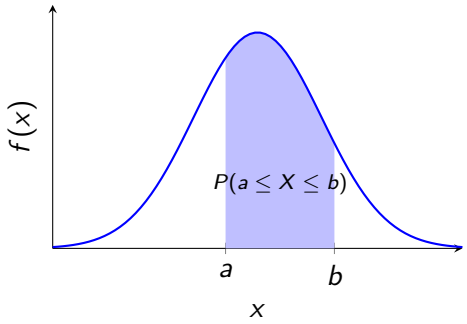
$$F(x) = \begin{cases} 0 & x < 0 \\ x & 0 \leq x \leq 1 \\ 1 & x > 1 \end{cases}$$

- Interval probabilities work exactly as before:

$$P(0.65 < X < 0.70) = F(0.70) - F(0.65) = 0.05.$$

Probability Density Function (PDF)

- Probability density function: The slope of the CDF,
$$f(x) = F'(x).$$
- Probabilities are areas under the PDF:
$$P(a \leq X \leq b) = \int_a^b f(x)\,dx = F(b) - F(a).$$
- $f(x) \geq 0$, and the total area under f is 1.
- $f(x)$ is a density, not a probability. It can be greater than 1. Only areas are probabilities.



Quantiles and Expected Values

- Quantile: The q quantile x_q solves $F(x_q) = q$. The median is $x_{0.5}$, and the 90th percentile is $x_{0.9}$. These are usually easier to find for continuous random variables.
- Expected value: The same idea as in Lecture 5, with the sum replaced by an integral,

$$E(X) = \int x\, f(x)\, dx.$$

- Variance: $\mathrm{Var}(X) = E[(X-\mu)^2] = E(X^2) - \mu^2$, as before.
- All the rules from Lecture 5 carry over: $E(a+bX) = a + b\,E(X)$, $\mathrm{Var}(a+bX) = b^2\,\mathrm{Var}(X)$, and standardization.
- You will not be asked to integrate. For each standard distribution, we report the mean and variance and use them.

Uniform Distribution

- $X \sim U(a, b)$: all values in $[a, b]$ are equally likely,

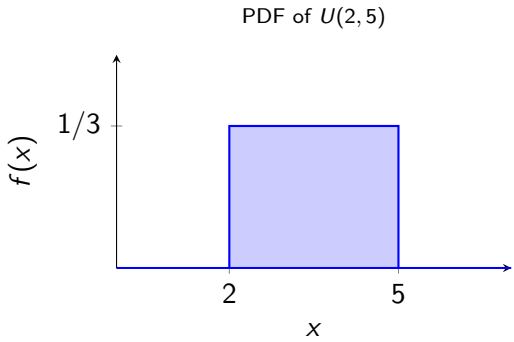
$$f(x) = \frac{1}{b-a} \quad \text{for } a \leq x \leq b,$$

and $f(x) = 0$ otherwise.

- CDF:

$$F(x) = \frac{x-a}{b-a} \quad \text{for } a \leq x \leq b.$$

- $U(0, 1)$ is called the standard uniform.



Uniform: Quantiles and Moments

- Quantiles: Invert the CDF,
$$x_q = a + q(b - a).$$
The median is the midpoint $(a + b)/2$.
- Mean and variance:
$$E(X) = \frac{a+b}{2}, \qquad \mathrm{Var}(X) = \frac{(b-a)^2}{12}.$$
- $X \sim U(2, 5)$: $E(X) = 3.5$, $\mathrm{Var}(X) = 9/12 = 0.75$, $\sigma \approx 0.866$, and the 90th percentile is $2 + 0.9 \times 3 = 4.7$.
- A linear function of a uniform is uniform: if $X \sim U(2, 5)$, then $2X + 2 \sim U(6, 12)$.

Uniform: An Example

A delivery is equally likely to arrive at any time between 9:00 and 11:00. Let X be the arrival time in minutes after 9:00, so $X \sim U(0, 120)$.

- Arrives before 9:30: $P(X < 30) = 30/120 = 0.25$.
- Arrives exactly at 10:00: $P(X = 60) = 0$. Only intervals have positive probability.
- Expected arrival: $E(X) = 60$, i.e., 10:00. Standard deviation: $120/\sqrt{12} \approx 34.6$ minutes.
- 90th percentile: $0.9 \times 120 = 108$, i.e., 10:48. You can be 90% sure the delivery arrives by then.
- The standard uniform is the building block of simulation: `RAND()` in Excel and `runif()` in R draw from $U(0, 1)$.

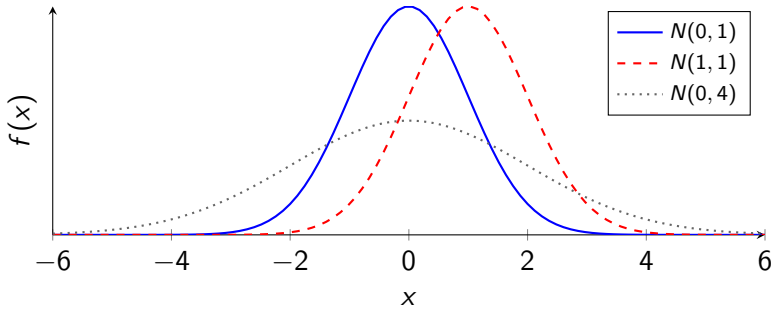
Normal Distribution

- $X \sim N(\mu, \sigma^2)$, with PDF

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \qquad -\infty < x < \infty.$$

- Mean and variance: $E(X) = \mu, \quad \mathrm{Var}(X) = \sigma^2$.
- Bell-shaped and symmetric around μ, so mean = median = mode. The total area is 1, with 0.5 on each side of μ.
- Also called the Gaussian distribution. $N(0, 1)$ is the standard normal.
- Why it is everywhere: sums and averages of many small independent shocks are approximately normal.

Normal Shapes



- Increasing μ shifts the curve to the right. Increasing σ flattens and spreads it out.
- Otherwise the shape is identical, so one table (the standard normal) is enough for all normals.

Linear Functions of a Normal

- Any random variable has $E(a+bX) = a + b\,E(X)$ and $\mathrm{Var}(a+bX) = b^2\,\mathrm{Var}(X)$.
- Normal random variables have an extra property: a linear function of a normal is also normal,
$$X \sim N(\mu, \sigma^2) \;\Rightarrow\; a + bX \sim N(a + b\mu,\; b^2\sigma^2).$$
- Example: $X \sim N(2, 9)$. Then
$$X + 1 \sim N(3, 9), \qquad 2X \sim N(4, 36), \qquad -2X \sim N(-4, 36).$$
- Nonlinear functions of a normal are generally not normal. For example, X^2 cannot be negative.

The Standard Normal

- Standardization is especially useful for normals:

$$X \sim N(\mu, \sigma^2) \;\Rightarrow\; Z = \frac{X - \mu}{\sigma} \sim N(0, 1).$$

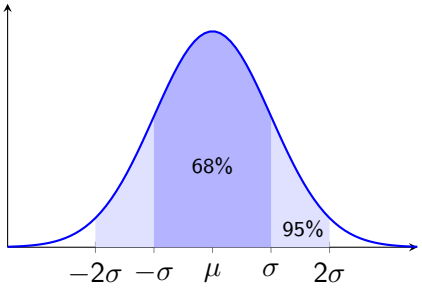
- $\Phi(z) = P(Z \leq z)$ denotes the standard normal CDF.
- Any normal probability can be written with Φ:

$$P(X \leq x) = P\left(Z \leq \frac{x - \mu}{\sigma}\right) = \Phi\left(\frac{x - \mu}{\sigma}\right).$$

- The z-score $(x - \mu)/\sigma$ measures how many standard deviations x is from the mean.
- Φ has no simple formula, but it is in every table and every software package.

The Empirical Rule

- For any normal distribution:
  - $P(\mu - \sigma < X < \mu + \sigma) \approx 0.68$
  - $P(\mu - 2\sigma < X < \mu + 2\sigma) \approx 0.95$
  - $P(\mu - 3\sigma < X < \mu + 3\sigma) \approx 0.997$
- This is where the 68–95–99.7 rule from Lecture 2 comes from.
- More precise values used throughout the course:
  - $P(|Z| \leq 1.645) = 0.90$
  - $P(|Z| \leq 1.96) = 0.95$
  - $P(|Z| \leq 2.576) = 0.99$



Finding Standard Normal Probabilities

- A standard normal table gives $\Phi(z) = P(Z \leq z)$. Example: $P(Z \leq 1.00) = 0.8413$ (row 1.0, column 0.00).
- Useful rules:
  - Complement: $P(Z > z) = 1 - \Phi(z)$.
  - Symmetry: $P(Z > z) = P(Z < -z)$, so $\Phi(-z) = 1 - \Phi(z)$.
  - Intervals: $P(a \leq Z \leq b) = \Phi(b) - \Phi(a)$.
- $P(-1.5 \leq Z \leq 2.0) = 0.9772 - 0.0668 = 0.9104$.
- $P(Z \geq 1.75) = 1 - 0.9599 = 0.0401$, or equivalently $P(Z \leq -1.75) = 0.0401$.
- Excel: `NORM.S.DIST(z,TRUE)` and `NORM.DIST(x,`μ`,`σ`,TRUE)`. R: `pnorm(z)` and `pnorm(x,mu,sigma)`.
- Sketch the area first. Most errors are set-up errors, not table errors.

Normal Probabilities: Weekly Revenue

A store's weekly revenue is $X \sim N(25000, 4000^2)$.

- Revenue below 20,000:

$$P(X < 20000) = P\left(Z < \frac{20000 - 25000}{4000}\right) = \Phi(-1.25) \approx 0.106.$$

- Revenue above 30,000: $z = 1.25$, so $P(X > 30000) = 1 - \Phi(1.25) \approx 0.106$. The same as below 20,000, by symmetry.
- Revenue between 22,000 and 28,000: $z = \pm 0.75$, so

$$P(22000 < X < 28000) = \Phi(0.75) - \Phi(-0.75) = 0.7734 - 0.2266 \approx 0.547.$$

- Excel: `NORM.DIST(20000,25000,4000,TRUE)`. R: `pnorm(20000,25000,4000)`.

Percentiles: Working Backwards

- Sometimes we know the probability and want the value. The q quantile of $X \sim N(\mu, \sigma^2)$ is

$$x_q = \mu + z_q \, \sigma,$$

  where z_q is the q quantile of the standard normal, $\Phi(z_q) = q$.
- 90th percentile of Z: $\Phi(1.28) \approx 0.90$, so $z_{0.90} \approx 1.28$.
- 90th percentile of $X \sim N(100, 15^2)$: $100 + 1.28 \times 15 = 119.2$.
- The middle 90% of a normal lies within $\mu \pm 1.645 \, \sigma$.
- Excel: `NORM.INV(q,`μ`,`σ`)`. R: `qnorm(q,mu,sigma)`.

Percentiles: Safety Stock

A store's weekly demand is $X \sim N(500, 60^2)$. A delivery arrives once a week, and the current policy stocks 550 units.

- Probability of running out:

$$P(X > 550) = 1 - \Phi\left(\frac{550 - 500}{60}\right) = 1 - \Phi(0.83) \approx 0.20.$$

  The store runs out about once every five weeks.

- To run out only 2% of the time, stock the 98th percentile of demand:

$$x_{0.98} = 500 + 2.054 \times 60 \approx 623 \text{ units.}$$

- The extra 73 units are the cost of reliability. Quantiles translate a service target into an inventory level.